2017 IEEE 56th Annual Conference on Decision and Control (CDC)
December 12-15, 2017, Melbourne, Australia

Solving Optimal Power Flow with non-Gaussian Uncertainties via Polynomial Chaos Expansion*

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Abstract— **The increasing penetration of renewable energy sources to power grids necessitates a structured consideration of uncertainties for optimal power flow problems. Modeling uncertainties via continuous random variables of finite variance we propose a tractable convex formulation of the uncertain optimal power flow problem. The uncertainties can be (non-)Gaussian, multivariate and/or correlated. We employ polynomial chaos expansion to rewrite the infinite-dimensional random-variable optimization problem as a finite-dimensional convex second-order cone program. This problem can be solved efficiently in a single numerical run for all realizations of the uncertainty. The solution provides a feedback policy in terms of the fluctuations. No Monte Carlo sampling is required to obtain either the solution or its statistics. The reduced computational effort and yet consistent results stemming from polynomial chaos are demonstrated in comparison to a Monte-Carlo-based solution for the IEEE 300-bus test system.**

Index Terms— **Optimal power flow, polynomial chaos, random-variable optimization, stochastic uncertainties**

I. Introduction

Solving the optimal power flow (OPF) problem is a standard task in power system operational planning. The objective is to minimize the operational cost subject to the physics-imposed power flow equations while respecting technical limitations, e.g. generation constraints, line flow limits. The steadily increasing influx of renewable energy sources to the power grid adds another problem dimension to OPF: the structured consideration of uncertainties and constraint satisfaction despite stochasticity.

Introducing chance constraints to the OPF problem is a possible remedy. The main challenge then is to recover tractable problem formulations. For the single-stage OPF problem under DC conditions (DC-OPF) subject to Gaussian uncertainties there exist several results: In [3, 11] individual chance constraints are employed for the operational planning problem. Therein, the Gaussian chance constraints are analytically reformulated leading to convex reformulations for the OPF problem. The authors of [9] build upon [3, 11], consider distributionally robust chance constraints, and introduce a modified cutting-plane algorithm.

In contrast to the extensive treatment of OPF with Gaussian uncertainties, the necessity to consider non-Gaussian, possibly multivariate and correlated uncertainties has been pointed out in the literature. For example, solar irradiance can be appropriately modeled via Beta and Dirichlet distributions [2, 15]. Also, the authors of [5] show that residential load patterns follow Beta, Gamma, or log-normal distributions. Wind power is modeled via Gaussian distributions in [3, 11], or entirely without any underlying distribution using Markov matrices [17, 18]. The latter works [17, 18] consider a joint chance-constrained formulation that is reformulated using a scenario-based approach. Whilst being able to consider non-Gaussian uncertainties and guarantee chance constraint feasibility, this approach typically yields conservative results. In order to satisfy the power flow equations for all realizations of the uncertainty, all above works consider power flow under nominal conditions as an equality constraint together with proportional automatic generation control accounting for any mismatches. Similar, but more general piece-wise affine policies have been considered in [13]. Therein, the feedback gains are treated as additional degrees of freedom of the optimization problem.

The contribution of the present work is the formulation of stochastic single-stage OPF problems in terms of random variables, allowing for non-Gaussian uncertainties. This leads to infinite-dimensional random-variable optimization problems for which we provide a tractable finite-dimensional formulation as a second-order cone program. The power flow equations are considered a function of random variables, yielding power balance for all realizations. We employ polynomial chaos expansion (PCE) for uncertainty propagation. All random fluctuations are then accounted for by the power injection because PCE parameterizes the solution in terms of the realization of the uncertainty—any uncertain demand is covered by the corresponding generation. We compute these generations for all realizations of the uncertainty in a single numerical run without having to sample.

The proposed problem formulation considers chance constraints as in [11, 13]. In contrast to [11, 13] we allow for non-Gaussian uncertainties and employ PCE. Applicability of PCE to (optimal) AC power flow has been demonstrated in [10]. For radial grids, the uncertain AC power flow can be efficiently solved by combining PCE with the backward-forward-sweep method [1]. In contrast to [1, 10], here we focus on DC conditions, provide a thorough treatment of satisfaction of power balance despite uncertainty, and show that PCE provides feedback policies.

II. Preliminaries

Consider a linear N-bus electrical network with lines $\mathcal{N}_l = \{1, \ldots, M\}$ under DC power flow conditions, i.e. small angle

* TM, TF, and VH acknowledge support of this work by the Helmholtz Association under the Joint Initiative "Energy System 2050 – A Contribution of the Research Field Energy". TF acknowledges further support from the Baden-Württemberg Stiftung under the Elite Programme for Postdocs.

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differences, no Ohmic losses, per-unit voltage magnitudes equal to one. The bus index set is $\mathcal{N} = \{1, \ldots, N\}$ with $N \in \mathbb{N}$; the set of all generation buses is $\mathcal{N}_g = \{n_1, \ldots, n_{N_g}\} \subseteq \mathcal{N}$ with $0 < N_g \leq N$; the slack bus is $i_s \in \mathcal{N}$; the net active power injection $p \in \mathbb{R}^N$ is decomposed as follows

$$p = d + Cu, \tag{1}$$

where d represents fixed power generation ($d_i > 0$) to or demand ($d_i < 0$) from the network. The vector $u \in \mathbb{R}^{N_g}$ represents the N_g degrees of freedom in the later OPF problem. The orthogonal matrix $C \in \mathbb{N}_0^{N \times N_g}$ is given as

$$C_{n_j,j} = \begin{cases} 1, & \forall\, n_j \in \mathcal{N}_g, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

In essence, C maps the free power injections u to their corresponding bus index. Under DC power flow total generation equals total demand as there are no Ohmic losses

$$\sum_{i=1}^{N} p_i = \mathbf{1}_{N_g}^\top u + \mathbf{1}_N^\top d = 0. \tag{3}$$

Also, DC power flow results in a linear dependence between the net active power injections p and the line power flows $p_l = \Phi p \in \mathbb{R}^M$ via the power transfer distribution factor (PTDF) matrix $\Phi \in \mathbb{R}^{M \times N}$. The PTDF matrix contains the network topology and branch susceptances [6].

Given a positive definite quadratic cost function, the deterministic DC-OPF problem with box constraints becomes

$$\begin{aligned}
\min_{u \in \mathbb{R}^{N_g}} \quad & \frac{1}{2} u^\top H u + h^\top u && \text{(4a)}\\
\text{s.t.} \quad & \mathbf{1}_{N_g}^\top u + \mathbf{1}_N^\top d = 0, && \text{(4b)}\\
& \underline{u}_i \leq u_i \leq \overline{u}_i, \quad i \in \mathcal{N}_g, && \text{(4c)}\\
& \underline{p}_{l,j} \leq \Phi_j (Cu + d) \leq \overline{p}_{l,j}, \quad j \in \mathcal{N}_l, && \text{(4d)}
\end{aligned}$$

with $H \in \mathbb{R}^{N_g \times N_g}$ positive definite. Problem (4) is a convex quadratic program (QP). Uncertainty is assumed to enter via the fixed power d. This may include stochasticity of the loads, uncertainty due to wind power, or solar infeed. As pointed out before, commonly used uncertainty descriptions include Gaussian, Beta, Gamma distributions, cf. [2, 5, 15]. For the scope of this work, we do not prefer one uncertainty model to another. We rather focus on what is common to all of them: *uncertainty is considered a random variable*, possibly non-Gaussian, multivariate and correlated.

Assumption 1 (Stochastic fixed power): The fixed power for bus $i = 1, \ldots, N$ is given by a second-order continuous random variable $\mathsf{d}_i \in \mathrm{L}^2(\Omega, \mu; \mathbb{R})$.[1] □

Under Assumption 1 we suggest to change the decision variable u in QP (4) to the random variable u, now an element of the Hilbert space of second-order vector-valued random variables $\mathrm{L}^2(\Omega, \mu; \mathbb{R})$, and solve the following problem

$$\begin{aligned}
\min_{\mathsf{u} \in \mathrm{L}^2(\Omega,\mu;\mathbb{R})} \quad & \mathrm{E}\left[\frac{1}{2} \mathsf{u}^\top H \mathsf{u} + h^\top \mathsf{u}\right] && \text{(5a)}\\
\text{s.t.} \quad & \mathbf{1}_{N_g}^\top \mathsf{u} + \mathbf{1}_N^\top \mathsf{d} = 0, && \text{(5b)}\\
& \underline{u}_i \leq \mathrm{E}[\mathsf{u}_i] \pm \beta_u \sqrt{\mathrm{Var}[\mathsf{u}_i]} \leq \overline{u}_i, && \text{(5c)}\\
& \underline{p}_{l,j} \leq \mathrm{E}\left[\phi_j^\top (C\mathsf{u} + \mathsf{d})\right] \\
& \qquad \pm \beta_l \sqrt{\mathrm{Var}\left[\phi_j^\top (C\mathsf{u} + \mathsf{d})\right]} \leq \overline{p}_{l,j}, && \text{(5d)}\\
& i \in \mathcal{N}_g, \; j \in \mathcal{N}_l. && \text{(5e)}
\end{aligned}$$

Here, $\mathrm{E}[\cdot]$ is the expected value, $\mathrm{Var}[\cdot]$ is the variance, u_i is the i-th entry of u, and $\phi_j^\top$ is the j-th row of the PTDF matrix Φ. The inequality constraints (5c), (5d) represent analytical reformulations of distributionally robust chance constraints with the factors β_l, β_u being functions of the acceptable violation probability of the original chance constraints and the assumptions about the distribution [4, 12, 14].[2]

Remark 1 (Random-variable OPF): Problem (5) differs from standard stochastic programming approaches to OPF, because the decision variable u is an element of the Hilbert space of second-order random variables, hence a measurable function, rendering Problem (5) infinite-dimensional. We call Problem (5) a *random-variable problem.* □

Remark 2 (Random-variable DC power flow): Assuming it exists, the optimal solution $\mathsf{u}^\star$ to (5) yields the distribution of the inputs parameterized by the fixed power d

$$\mathsf{u}^\star = \mathsf{u}^\star(\mathsf{d}). \tag{6}$$

By construction of (5), it is ensured that total demand equals total generation, as the DC power flow (4b) is written as a (linear) function of random variables (5b). It *is not* treated as a chance constraint. For *any* realization d of d the according realization $u^\star(d)$ of $\mathsf{u}^\star(\mathsf{d})$ satisfies the deterministic DC power flow $\mathbf{1}_{N_g}^\top u^\star(d) + \mathbf{1}_N^\top d = 0$. In that sense the solution (6) to Problem (5) is a feedback policy that is computed *before* the actual realization of d is known; it is computed solely based on the stochasticity of d. Without any need for further online-optimization the power set points are obtained, which are optimal in the sense of Problem (5). □

Note that the random-variable DC power flow allows for more general uncertainty considerations than existing approaches to DC power flow subject to uncertainty. For example, the authors of [17] formulate the energy balance for the nominal case, and let automatic generation control account for any power mismatch via proportional feedback. In the case of random-variable DC power flow, also higher-order moments of demand and generation are matched.

We will employ polynomial chaos expansion (PCE) to solve (5), because it natively copes with non-Gaussian uncertainties d. Furthermore, PCE provides a tractable deterministic formulation of Problem (5) that can be solved in just a single numerical run.

[1] Accurately, given the probability space $(\Omega, \sigma(\Omega), \mu)$ the elements of $\mathrm{L}^2(\Omega, \mu; \mathbb{R})$ are equivalence classes under the equivalence relation $\mathsf{x} \equiv \mathsf{y} \Leftrightarrow \mathsf{x} = \mathsf{y}$ almost everywhere for $\mathsf{x}, \mathsf{y} \in \mathrm{L}^2(\Omega, \mu; \mathbb{R})$. For sake of readability, we do not distinguish between the equivalence class and its representatives. Also, the support of μ is assumed to coincide with the sample space Ω.

[2] Here, due to space limitations they are considered as being already analytically reformulated.

III. Polynomial Chaos Expansion

The following introduction to PCE is non-exhaustive. For further details see e.g. [16, 19]. Let $\xi = [\xi_1, \ldots, \xi_{n_\xi}]^\top$ be an $\mathbb{R}^{n_\xi}$-valued random vector, called *stochastic germ*. Each component ξ_i is a second-order random variable from the Hilbert space $\mathrm{L}^2(\Omega_i, \mu_i; \mathbb{R})$ for all $i = 1, \ldots, n_\xi$. Following [16], the L^2 space over the product probability space is

$$\mathrm{L}^2(\Omega, \mu; \mathbb{R}) = \bigotimes_{i=1}^{n_\xi} \mathrm{L}^2(\Omega_i, \mu_i; \mathbb{R}), \tag{7}$$

where $\Omega = \Omega_1 \times \cdots \times \Omega_{n_\xi}$, and $\mu = \mu_1 \otimes \cdots \otimes \mu_{n_\xi}$. Polynomial chaos allows rewriting second-order random variables $\mathsf{x} \in \mathrm{L}^2(\Omega, \mu; \mathbb{R})$ as

$$\mathsf{x} = \sum_{l=0}^{\infty} x_l \psi_l \text{ with } x_l = \frac{\langle \mathsf{x}, \psi_l \rangle}{\langle \psi_l, \psi_l \rangle} \in \mathbb{R}, \tag{8}$$

where the basis $\{\psi_l\}_{l=0}^{\infty}$ spans $\mathrm{L}^2(\Omega, \mu; \mathbb{R})$ and is orthogonal[3], i.e. for all $l, k \in \mathbb{N}_0$

$$\mathrm{E}[\psi_l \psi_k] = \langle \psi_l, \psi_k \rangle = \int_\Omega \psi_l(\tau) \psi_k(\tau) \mathrm{d}\mu(\tau) = \gamma_l \delta_{lk}, \tag{9}$$

where γ_l is a positive constant, and δ_{lk} is the Kronecker-delta. W.l.o.g. we choose $\psi_0 = 1$. The real-valued x_l in (8) are called PCE coefficients. Viable univariate bases matching the uncertainty are given by the Askey-scheme [19], see Table I. For practical applications the infinite sum (8) is truncated after $L+1 \in \mathbb{N}$ terms,

$$\mathsf{x} \approx \hat{\mathsf{x}} := \sum_{l=0}^{L} x_l \psi_l \text{ with } x_l = \frac{\langle \mathsf{x}, \psi_l \rangle}{\langle \psi_l, \psi_l \rangle} \in \mathbb{R}. \tag{10}$$

The truncation error $\|\mathsf{x} - \hat{\mathsf{x}}\|$ is orthogonal to the subspace $\operatorname{span}\{\psi_l\}_{l=0}^L \subseteq \mathrm{L}^2(\Omega, \mu; \mathbb{R})$ and L^2-optimal, i.e. $\lim_{L \to \infty} \|\mathsf{x} - \hat{\mathsf{x}}\| = 0$, in the induced norm $\|\cdot\|$, cf. [16, 19]. If the bases of $\mathrm{L}^2(\Omega_i, \mu_i; \mathbb{R})$ have dimension $n_d + 1 \in \mathbb{N}$, the dimension of the subspace $\operatorname{span}\{\psi_l\}_{l=0}^L$ is

$$L+1 = \frac{(n_\xi + n_d)!}{n_\xi! n_d!}. \tag{11}$$

The subspace is spanned by orthogonal basis polynomials in n_ξ variables of degree at most n_d. Note that for $\mathbb{R}^n$-valued random vectors PCE is equally applicable. The PCE coefficients x_l are then vector-valued $x_l \in \mathbb{R}^n$ with each component satisfying the Fourier formula (10). Properties of PCE entail [16, Chapter 11]:

- P1 Non-Gaussian random variables can be considered; specifically, non-symmetric multivariate correlated random variables with finite support.
- P2 The m-th stochastic moment is a polynomial of degree m in the $L+1$ PCE coefficients. For example, $\mathrm{E}[\hat{\mathsf{x}}] = x_0$, and $\mathrm{Var}[\hat{\mathsf{x}}] = \boldsymbol{x}^\top W \boldsymbol{x}$, where $\boldsymbol{x} = [x_0 \ \ldots \ x_L]^\top \in \mathbb{R}^{L+1}$ is the vector of PCE coefficients and the positive semidefinite matrix W is the Gramian with the first entry set to zero:
$$W = \operatorname{diag}(0, \langle \psi_1, \psi_1 \rangle, \ldots, \langle \psi_L, \psi_L \rangle). \tag{12}$$
- P3 Using Galerkin projection, PCE allows to reformulate stochastic problems as deterministic problems.
- P4 Sampling is computationally cheap, because ξ is composed of standard random variables.

[3] The basis functions ψ_i are dichotomous: they are both a random variable and a polynomial function [16, Remark 11.15]. Whenever we wish to emphasize the former over the latter the argument ξ is dropped.

TABLE I

Askey-scheme for 2^{nd}-order continuous random variables.

Type	Support	Polynomial Basis
Beta	$[-1, 1]$	Jacobi
Gamma	$[0, \infty)$	Laguerre
Gaussian	$(-\infty, \infty)$	Hermite
Uniform	$[-1, 1]$	Legendre

Note that the dimension n_ξ of the stochastic germ ξ stands for the number of drivers of uncertainty. In power system applications the *number of uncertain injections* is often large, e.g. many renewable energy production units, whilst the *number of drivers of uncertainty* are less numerous, e.g. forecast errors in wind are driven by the same forecast uncertainty in wind speeds. This is a modelling choice.

IV. Random-variable DC-OPF Using PCE

Next, we show that the properties of PCE allow an exact reformulation of Problem (5) as a deterministic finite-dimensional second-order cone program. First, Assumption 1 is made more precise in the light of PCE. For sake of readability the $\hat{\mathsf{x}}$-notation from (10) is dropped, i.e. $\mathsf{x} \leftarrow \hat{\mathsf{x}}$.

Assumption 2 (Uncertain fixed power using PCE*):* For $i = 1, \ldots, N$, the uncertain fixed power d is modeled as the realization of the known $\mathbb{R}^N$-valued second-order random vector d with entries $\mathsf{d}_i \in \mathrm{D} \subset \mathrm{L}^2(\Omega, \mu; \mathbb{R})$. The PCE coefficients $d_l \in \mathbb{R}^N$ are assumed known w.r.t. the polynomial basis $\{\psi_l\}_{l=0}^L$ such that $\mathrm{D} = \operatorname{span}\{\psi_l\}_{l=0}^L$. □

Note that Assumption 2 implies Assumption 1. We introduce PCE for the uncertain fixed power d according to Assumption 2, and introduce PCE for the input u as follows

$$\mathsf{d} = \sum_{i=0}^{L} d_i \psi_i = \boldsymbol{D}\boldsymbol{\psi}, \quad \mathsf{u} = \sum_{i=0}^{L} u_i \psi_i = \boldsymbol{U}\boldsymbol{\psi}, \tag{13a}$$

$$\text{with} \quad \boldsymbol{\psi} = [\psi_0 \quad \psi_1 \quad \ldots \quad \psi_L]^\top, \tag{13b}$$

$$\boldsymbol{D} = [d_0 \quad d_1 \quad \ldots \quad d_L] \in \mathbb{R}^{N \times (L+1)}, \tag{13c}$$

$$\boldsymbol{U} = [u_0 \quad u_1 \quad \ldots \quad u_L] \in \mathbb{R}^{N_g \times (L+1)}. \tag{13d}$$

The PCE coefficients $\boldsymbol{U}$ parameterize the generation power in terms of the realization of the uncertainty. Denoting $\boldsymbol{u} = \operatorname{vec}(\boldsymbol{U}) \in \mathbb{R}^{N_g(L+1)}$, the cost function (5a) becomes

$$J(\boldsymbol{u}) = \frac{1}{2} u_0^\top H u_0 + h^\top u_0 + \frac{1}{2} \boldsymbol{u}^\top (W \otimes H) \boldsymbol{u}, \tag{14}$$

cf. P2. The convex quadratic cost function (14) is a reformulation of (5a) in the PCE coefficients of u. The first two terms in (14) cover the cost of nominal/expected demand, while the third term accounts for the cost of fluctuations in the fixed power. Similar to the cost function (5a), the energy balance (5b) can be rewritten in terms of the PCE coefficients of u using Galerkin projection for uncertainty propagation

$$0=\Big\langle \mathbf{1}_{N_g}^\top \sum_{i=0}^{L} u_i\psi_i+\mathbf{1}_N^\top \sum_{i=0}^{L} d_i\psi_i, \psi_k\Big\rangle,\; k=0,\dots,L \quad (15a)$$

which is equivalent to

$$(I_{L+1}\otimes \mathbf{1}_{N_g}^\top)\boldsymbol{u}+(I_{L+1}\otimes \mathbf{1}_N^\top)\boldsymbol{d}=0, \quad (15b)$$

with $\boldsymbol{d}=\operatorname{vec}(\boldsymbol{D})\in\mathbb{R}^{N(L+1)}$. Note that this way the random-variable DC power flow (5b) is rewritten in terms of the PCE coefficients $\boldsymbol{d}\in\mathbb{R}^{N(L+1)}$ and $\boldsymbol{u}\in\mathbb{R}^{N_g(L+1)}$. In other words, propagating the uncertainty d through the power flow equations is equivalent to solving a system of (linear) equations in terms of the PCE coefficients, cf. P3.

Lastly, the inequality constraints (4c), (4d) are rewritten as second-order cone constraints using P2. To this end, consider a generic real-valued random variable x with PCE coefficients $\boldsymbol{x}=[x_0\ x_1\ \dots\ x_L]^\top$, and a generic positive β

$$\mathrm{E}[\mathsf{x}]\pm\beta\sqrt{\mathrm{Var}[\mathsf{x}]}=w^\top\boldsymbol{x}\pm\beta\big\|W^{\frac{1}{2}}\boldsymbol{x}\big\|_2, \quad (16)$$

with $w=[1\,0\,\dots\,0]^\top\in\mathbb{R}^{L+1}$.

Remark 3 (Higher-order moments): If higher-order moments are added to the constraints (5c), (5d), the PCE property P2 provides deterministic reformulations thereof. For example, consider for some $\nu\in\mathbb{R}$

$$\underline{x}\leq\mathrm{E}[\mathsf{x}]\pm\beta\sqrt{\mathrm{Var}[\mathsf{x}]}\pm\nu s\leq\overline{x}, \quad (17)$$

where $s=\mathrm{E}\left[((\mathsf{x}-\mu)/\sigma)^3\right]$ is the skewness of the random variable x with mean μ and standard deviation σ. Using PCE w.r.t. x the third-order moment $\mathrm{E}\left[\mathsf{x}^3\right]$ becomes

$$\mathrm{E}\left[\mathsf{x}^3\right]=\sum_{i,j,k=0}^{L}x_ix_jx_k\langle\psi_i\psi_j,\psi_k\rangle, \quad (18)$$

from which the skewness is computed as

$$s=s(\boldsymbol{x})=\frac{\mathrm{E}\left[\mathsf{x}^3\right]-3\mu\sigma^2-\mu^3}{\sigma^3}, \quad (19)$$

with $\mu=x_0$, and $\sigma^2=\boldsymbol{x}^\top W\boldsymbol{x}$, cf. P2. Of course, the constraint is no longer a second-order cone constraint. □

Let $\hat{u}_i^\top$, $\hat{p}_{l,j}^\top$ denote the i-th, j-th row of $\boldsymbol{U}$, $\Phi(C\boldsymbol{U}+\boldsymbol{D})$, respectively. Using the above formulations (14)–(16), the uncertain DC-OPF Problem (5) can be written as a second-order cone program (SOCP) upon introducing a new decision variable $v\in\mathbb{R}$

$$\min_{\boldsymbol{u},v}\quad v \quad (20a)$$

$$\text{s.t.}\quad (I_{L+1}\otimes\mathbf{1}_{N_g}^\top)\boldsymbol{u}+(I_{L+1}\otimes\mathbf{1}_N^\top)\boldsymbol{d}=0, \quad (20b)$$

$$\Big\|\frac{1}{\sqrt{2}}\big(\mathbf{H}^{\frac{1}{2}}\boldsymbol{u}+\mathbf{H}^{-\frac{1}{2}}\mathbf{h}\big)\Big\|_2\leq v, \quad (20c)$$

$$\underline{u}_i\leq w^\top\hat{u}_i\pm\beta_u\big\|W^{\frac{1}{2}}\hat{u}_i\big\|_2\leq\overline{u}_i, \quad (20d)$$

$$\underline{p}_{l,j}\leq w^\top\hat{p}_{l,j}\pm\beta_l\big\|W^{\frac{1}{2}}\hat{p}_{l,j}\big\|_2\leq\overline{p}_{l,j}, \quad (20e)$$

$$i\in\mathcal{N}_g,\ j\in\mathcal{N}_l, \quad (20f)$$

cf. [7], where $\mathbf{H}=\operatorname{diag}(1,\langle\psi_1,\psi_1\rangle,\dots,\langle\psi_L,\psi_L\rangle)\otimes H\succ 0$, $\mathbf{h}=w\otimes h$. The solution of (20) entails the optimal PCE coefficients, from which statistics of the optimal input u can be inferred. Importantly, the PCE coefficients specify the feedback policy, i.e. how generation is adjusted *upon* measuring the realization of the uncertainty. The solution of (20) is related to the solution of the original random-variable problem (5) as follows.

Proposition 1 (Correspondence of solutions): Let Assumption 2 hold, let the optimal solution of the random-variable problem (5) exist and be $\mathsf{u}^\star\in\mathrm{L}^2(\Omega,\mu;\mathbb{R})$, and let $\boldsymbol{u}^\star=\operatorname{vec}(\boldsymbol{U}^\star)\in\mathbb{R}^{N_g(L+1)}$ be the minimizer of the SOCP (20) with respect to $\boldsymbol{u}$. Then, the solution $\mathsf{u}^\star$ satisfies $\mathsf{u}^\star=\boldsymbol{U}^\star\boldsymbol{\psi}$. □

Proof: First, note that the random-variable DC power flow (4b) is linear. Because the fixed power uncertainty d allows an exact PCE in $L+1$ dimensions, cf. Assumption 2, no truncation error is introduced when choosing the same dimension for u in (13). Second, note that all PCE-based reformulations are mere substitutions and algebraic manipulations, i.e. the cost function reformulation (14) and the constraint reformulation (16). Consequently, Problem (20) is equivalent to Problem (5) w.r.t. the solution $\boldsymbol{u}$. ■

Note that Proposition 1 ensures a posteriori that (5) can be reformulated as a finite-dimensional convex problem under the given Assumptions. In light of Remarks 1 and 2 the solution $\boldsymbol{u}^\star$ of the SOCP (20) deserves further attention.

Corollary 1 (Properties of $\boldsymbol{u}^\star$): Let SOCP (20) admit an optimal solution $\boldsymbol{u}^\star$, then the following holds:

(i) The optimal random variable is $\mathsf{u}^\star=\boldsymbol{U}^\star\boldsymbol{\psi}$ with $\boldsymbol{u}^\star=\operatorname{vec}(\boldsymbol{U}^\star)$, more precisely $\mathsf{u}^\star=\mathsf{u}^\star(\mathsf{d})=\boldsymbol{U}^\star(\boldsymbol{D})\boldsymbol{\psi}$, cf. Remark 2.

(ii) Given the realization d of d, the corresponding optimal realization w.r.t. power generation (in the sense of Problem (5)) is immediately obtained as $u^\star=u^\star(d)$.

(iii) The cost attains its minimum in terms of the expected value of the random variable $\mathsf{J}(\mathsf{u}^\star)=\frac{1}{2}\mathsf{u}^{\star\top}H\mathsf{u}^\star+h^\top\mathsf{u}^\star$.

(iv) The random-variable power flow (4b) is satisfied $\mathbf{1}_{N_g}^\top\mathsf{u}^\star(\mathsf{d})+\mathbf{1}_N^\top\mathsf{d}=0$. Consequently, it is satisfied for all realizations d of d as follows $\mathbf{1}_{N_g}^\top u^\star(d)+\mathbf{1}_N^\top d=0$.

(v) If the factors β_l,β_u are chosen as in [4, 12], the generation and line flow constraints admit a probabilistic guarantee in terms of chance constraints. □

To summarize, PCE allows an equivalent formulation (in terms of u) of the infinite-dimensional uncertain DC-OPF problem (5) as a tractable finite-dimensional SOCP (20). The type of uncertainty is specified via the PCE coefficients of the uncertain fixed power d, treated as a vector-valued random variable. It can be non-Gaussian, multivariate and possibly have correlated entries. Consequently, it allows for a broad and flexible uncertainty modelling. The power flow equations are always satisfied.

V. Simulation Example

The IEEE 300-bus test system serves as a demonstrator for stochastic DC-OPF using PCE. It has $N=300$ buses,

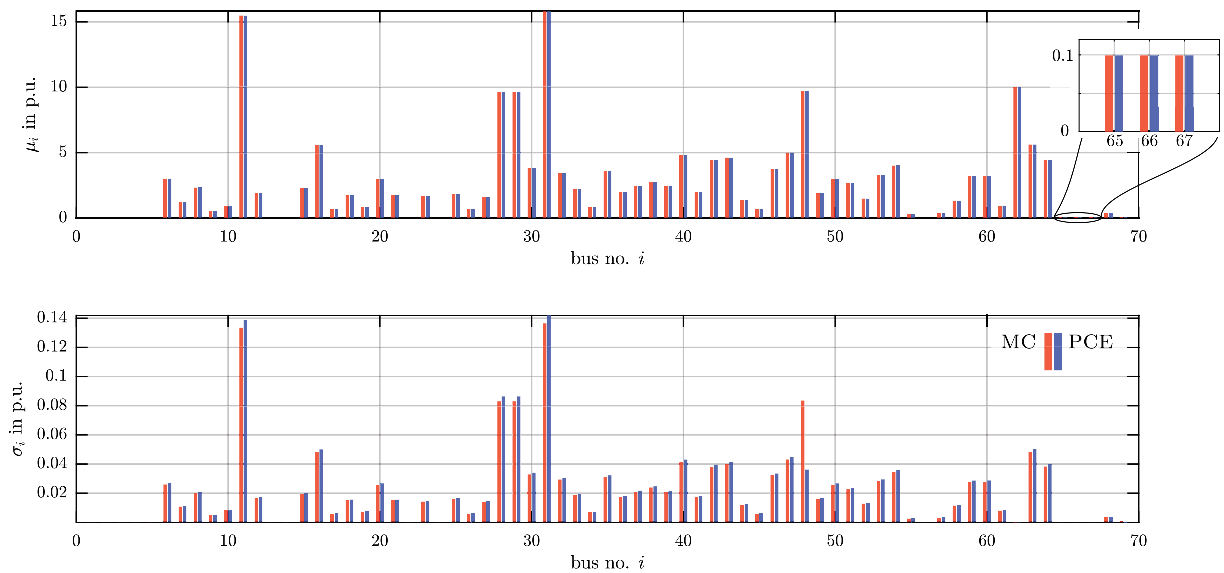


Fig. 1. Mean and standard deviation of generator buses $i \in \mathcal{N}_g$ from MC and PCE solution.

of which $N_g = 69$ are generators, and $M = 411$ lines. All deterministic/nominal values are taken from the Power Systems Test Case Archive.[4] All figures are given in per-unit.

The purpose of the simulation example is twofold: (i) demonstrate how the infinte-dimensional stochastic OPF problem (5) is tractably solved via its finite-dimensional PCE-equivalent (20), and (ii) compare the quality of the PCE solution to the hypothetical case of *full a posteriori optimization*, i.e. Monte Carlo simulation where a separate, deterministic OPF is run for each sample.

A. Uncertainty and Constraint Modelling

For our purposes we introduce the following sources of uncertainty: ξ_{d} represents the N_{d}-variate stochastic germ for the power demand with entries $\xi_{\mathrm{d},i}$ for $i = 1, \ldots, N_d$ and $N_d = 8 \leq N$, ξ_{s} the univariate stochastic germ for solar power, ξ_{w} the univariate stochastic germ for wind power. Thus, the $(2 + N_{\mathrm{d}})$-variate stochastic germ becomes $\xi = [\xi_{\mathrm{d}} \;\; \xi_{\mathrm{s}} \;\; \xi_{\mathrm{w}}]^\top$ and contains all sources of uncertainty. Table II summarizes the specific choice of distributions for the elements of ξ. For simplicity let each univariate basis $\{\phi_{\mathrm{s},k}\}_{k=0}^d$, $\{\phi_{\mathrm{w},k}\}_{k=0}^d$, $\{\phi_{\mathrm{d},i,k}\}_{k=0}^d$ have dimension $d+1 = 2$. Then, the dimension of the tensorized Hilbert space (7) is, $L+1 = 2+N_{\mathrm{d}}$, cf. (11), with basis $\{\psi_k\}_{k=0}^L$. We assume that solar/wind uncertainty affects every bus. Demand uncertainty affects selected buses $i \in \mathcal{N}_{\mathrm{u}}$ with

$$\mathcal{N}_u = \{231, 234, 235, 225, 228, 217, 218, 246\} \subseteq \mathcal{N}, \quad (21)$$

with $\mathcal{N} = \{1, \ldots, N\}$, and $|\mathcal{N}_{\mathrm{u}}| = N_d$ (using sequential bus numbering). For each bus $i \in \mathcal{N}$ the fixed yet uncertain active power d_i is modeled as the sum of wind power, solar power, and demand,

$$\mathsf{d}_i = \begin{cases} -\mathsf{p}_{\mathrm{d},i} + \mathsf{p}_{\mathrm{s},i} + \mathsf{p}_{\mathrm{w},i}, & \forall i \in \mathcal{N}_{\mathrm{u}} \\ -\hat{p}_{\mathrm{d},i} + \mathsf{p}_{\mathrm{s},i} + \mathsf{p}_{\mathrm{w},i}, & \forall i \in \mathcal{N} \setminus \mathcal{N}_{\mathrm{u}}, \end{cases} \quad (22a)$$

with

$$\mathsf{p}_{\mathrm{a},i} = \sum_{k=0}^{L} p_{\mathrm{a},ik} \psi_k(\xi), \quad \mathrm{a} \in \{\mathrm{s}, \mathrm{w}, \mathrm{d}\}. \quad (22b)$$

Consider the interval $\mathcal{I} = [0.85\, \hat{p}_{\mathrm{d},i}, 1.15\, \hat{p}_{\mathrm{d},i}]$, where $\hat{p}_{\mathrm{d},i}$ is the nominal power of bus i. Then, we choose the PCE coefficients $p_{\mathrm{d},ik}$ as follows: for Hermite stochastic germ (corresponding to a normal distribution, see Table I), set $\mathcal{I}$ equal to the 3σ-interval $[\mu_i - 3\sigma_i, \mu_i + 3\sigma_i]$; for Jacobi stochastic germ (corresponding to a Beta distribution, see Table I), set $\mathcal{I}$ equal to the support. The nominal solar power $\hat{p}_{\mathrm{s},i} = \alpha_{\mathrm{s},i}\, \hat{p}_{\mathrm{d},i}$ and nominal wind power $\hat{p}_{\mathrm{w},i} = \alpha_{\mathrm{w},i}\, \hat{p}_{\mathrm{d},i}$ are chosen based on the nominal power demand $\hat{p}_{\mathrm{d},i}$.[5] We set $\alpha_{\mathrm{s},i} = \alpha_{\mathrm{w},i} = 0.1$. The respective PCE coefficients $p_{\mathrm{s},ik}, p_{\mathrm{w},ik}$ are then computed analogously to $p_{\mathrm{d},ik}$. To enforce some constraints to be active, the lower generation limits for generators $\{65, 66, 67\}$ (buses $\{267, 292, 294\}$) are set to 0.1, and the upper limit for generator 62 (bus 263) is set to 10. Additionally, the limit for branch 394 (connecting buses 249/generator 48 and bus 3) is reduced to 9.95 p.u.. For simplicity, choose equal violation probabilities in (5c),

TABLE II
STOCHASTIC GERM FOR SIMULATION EXAMPLE, $i = 1, \ldots, 4$.

$\xi_{\mathrm{d},i}$	$\xi_{\mathrm{d},5}$	$\xi_{\mathrm{d},6}$	$\xi_{\mathrm{d},7}$	$\xi_{\mathrm{d},8}$	ξ_{s}	ξ_{w}
$\mathsf{N}(0,1)$	$\mathsf{B}(8,3)$	$\mathsf{B}(3,8)$	$\mathsf{B}(7,7)$	$\mathsf{B}(3,8)$	$\mathsf{B}(7,7)$	$\mathsf{N}(0,1)$

[4] Curated by University of Washington, Electrical Engineering, www.ee.washington.edu/research/pstca, visited October 17, 2016. See website for single-line diagram; not shown due to space limitations.

[5] We demand $\alpha_{\mathrm{s},i}, \alpha_{\mathrm{w},i} > 0$, and $\alpha_{\mathrm{s},i} + \alpha_{\mathrm{w},i} \leq \alpha_i < 1$, where α_i is chosen sufficiently small such that the realization of the fixed active power d_i is negative for sufficiently many realizations. This avoids net solar/wind feed-in that is not penalized in the cost function of the OPF problem.

i.e. (5d) equal: $\beta_u = \beta_l = \beta$. Specifically, according to [4] choose $\beta = \sqrt{(1-\varepsilon)/\varepsilon}$ with violation probability $\varepsilon = 2.5\,\%$. This choice corresponds to a (possibly conservative) distributionally robust reformulation that guarantees chance constraint satisfaction f-or any distribution with this specific mean and covariance [4].

The SOCP (20) is solved using yalmip [8] with Gurobi. The PCE-solution is obtained consistently in about 40 ms on a standard desktop computer, excluding pre- and post-processing. Hence, computational tractability of the SOCP and thus computational tractability of (5) is given.

B. A posteriori Optimality vs. Polynomial Chaos

The PCE-solution from (20) is compared to the Monte Carlo solution (MC) for a total of $N_{\text{MC}} = 10\,000$ samples of the stochastic germ ξ. The MC solution is 'true' in the sense of *full information*: for a total of N_{MC} realizations of the uncertain fixed power d the corresponding deterministic DC-OPF (4) is solved (using Matpower [20] with Gurobi). The N_{MC} solutions from MC provide the best possible distribution of random variable power generation, because for each realization the constraints are satisfied. Arguably, this also yields the best possible distribution of per-sample-optimal costs. Note how the PCE solution from (5a) differs from the solutions obtained through Monte Carlo: instead of N_{MC} problems the PCE solves just one optimization problem. Different from the Monte Carlo simulation, where any single solution might be vulnerable to variations in the random variables, the PCE solution is robust to uncertain variations. Further, the PCE solution specifies a feedback policy for all a priori unknown realizations, which will lead to a feasible solution with probability $1-\varepsilon$ as specified by the chance constrained formulation (5) via $\beta_a = \beta_a(1-\varepsilon)$ with $a \in \{u, l\}$. Since the violation probability ε is non-zero, there may exist realizations where the feedback policy leads to violations of the inequality constraints. To compare the quality of the PCE solution to the MC solution we compare statistics of the obtained solutions.

C. Numerical Results

Figure 1 shows the mean and standard deviation of all generator buses $i \in \mathcal{N}_g$ as obtained from MC and PCE. For MC the lower limit of 0.1 for the generators $\{65, 66, 67\}$ is attained for each sample, i.e. zero standard deviation; the same holds in this case for PCE as can be seen from the zoomed-in plot in Figure 1. The considered chance constraint is effectively a deterministic constraint in this case as the standard deviation is zero for PCE too. In terms of the PCE coefficients of the respective buses this means that any non-zero-order PCE coefficient is zero. While the mean of the power generation is equal for MC and PCE

$$\|\mu^{\text{MC}}\|_1 = \|\mu^{\text{PCE}}\|_1, \tag{23}$$

the power fluctuation is slightly bigger for PCE

$$\|\sigma^{\text{PCE}}\|_1 = \|\sigma^{\text{MC}}\|_1 + 0.0067. \tag{24}$$

That PCE yields a slightly different power fluctuation is related to solving the stochastic OPF problem in a single numerical run: due to the chance constraints the PCE solution has to assign the power fluctuations to different buses simultaneously. The MC solution, instead, finds the optimal configuration of power generation for each sample. Figure 1 shows that except for bus 48 the standard deviations from PCE are greater than or equal to the standard deviations from MC. To gain more insight, Figure 2a shows the PDF of power generation at generator 43 (bus 220): the difference between the PDF obtained from MC and PCE is small—with a slightly higher standard deviation for PCE ($\sigma_{43}^{\text{MC}} = 0.0398 < \sigma_{43}^{\text{PCE}} = 0.0413$). This is is to be expected from column 43 in Figure 1. The largest differences in mean and standard deviation occur directly at bus 48

$$\|\mu^{\text{MC}} - \mu^{\text{PCE}}\|_\infty = |\mu_{48}^{\text{MC}} - \mu_{48}^{\text{PCE}}| = 0.0068, \tag{25a}$$
$$\|\sigma^{\text{MC}} - \sigma^{\text{PCE}}\|_\infty = |\sigma_{48}^{\text{MC}} - \sigma_{48}^{\text{PCE}}| = 0.0473. \tag{25b}$$

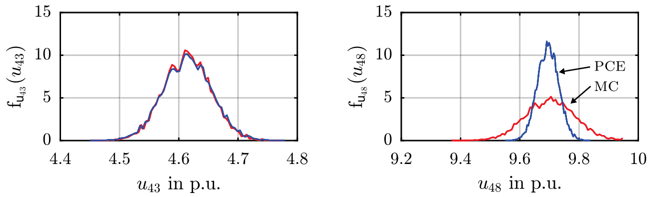
Figure 2b shows the PDF for generator 48 (bus 249). It manifests the significantly smaller standard deviation of the PDF from PCE ($\sigma_{43}^{\text{MC}} = 0.0835 > \sigma_{43}^{\text{PCE}} = 0.0362$), cf. column 48 from Figure 1. This behavior for generator 48 is attributed to the line flow limit of branch 394. The cumulative distribution function (CDF) of the power across line 394 for different values of $\beta_l = \sqrt{(1-\varepsilon)/\varepsilon}$ is shown in Figure Figure 3a, where the upper power flow limit is 9.95 p.u.. For MC there exist samples such that the upper constraint 9.95 is active. For PCE the (conservative) chance constraint reformulation introduces a safety margin that leads to the solution staying away from the upper limit. Lowering the violation probability ε leads to increasing the factor β_l, which leads to a steeper CDF and thus reduced fluctuation. In other words, PCE enforces reduced fluctuation in the vicinity of active constraints.

As shown above, the conservatism due to chance constraints forces PCE to choose a distribution of line flows and power generation that is different from the Monte Carlo solution. Therefore, the distribution of cost differs accordingly. However, as seen in Figure 3b, the cost distribution due to PCE comes very close to the a posteriori optimal cost distribution of the MC solution. In fact, the solution via PCE yields only slightly increased costs

$$\mu_{\text{J}}^{\text{PCE}} = \mu_{\text{J}}^{\text{MC}} + 80.0638\$/\text{hr} = 5.2753 \cdot 10^5\$/\text{hr} \tag{26a}$$
$$\sigma_{\text{J}}^{\text{PCE}} = \sigma_{\text{J}}^{\text{MC}} + \ \ 1.0180\$/\text{hr} = 5.5303 \cdot 10^3\$/\text{hr}. \tag{26b}$$

Obtaining the MC solution takes slightly more than 3 min. Of course, more advanced sampling techniques (Latin hypercube, Markov chain MC, etc.) can be applied. The PCE solution is thus obtained faster (40 ms) whilst not sacrificing quality of the obtained solution.

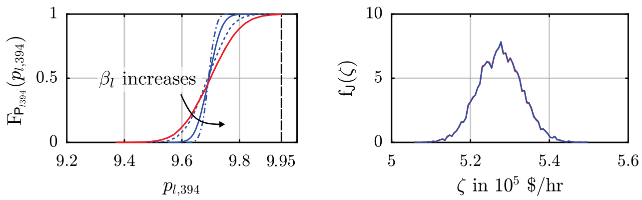
Remark 4 (PDFs via PCE): Using PCE, the PDFs are obtained as follows, cf. P4: to generate PCE samples evaluate the basis polynomials for each of the N_{MC} samples, multiply by the PCE coefficients, and sum up. We emphasize that for PCE this step is entirely post-processing and not required to obtain the solution to the SOCP (20). For MC, however, sampling is indispensable to obtain the solution. □



(a) Generator 43.

(b) Generator 48.

Fig. 2. Comparison between PDFs for active power obtained via MC (red) and PCE (blue).



(a) MC-CDF vs. PCE-CDF for security levels $\varepsilon \in \{0.1, 0.025, 0.01\}$ with $\beta_l = \sqrt{(1-\varepsilon)/\varepsilon}$.

(b) MC-PDF vs. PCE-PDF.

Fig. 3. CDF of power across line 394 and PDF of optimal cost function.

VI. Conclusion & Outlook

This work formulates the OPF problem subject to non-Gaussian multivariate uncertainties entirely in terms of random variables such that the DC power flow equations are always satisfied. Using PCE the problem is reformulated as an SOCP with PCE coefficients as decision variables that is solved a single numerical run. The optimal solution admits different interpretations: it yields the coefficients of power generation in terms of a polynomial basis of standard random variables; it yields a feedback policy in terms of the power fluctuation, computed only based on the stochasticity of the uncertainty. Immediately upon knowing the realization of the fluctuation the corresponding power injection is known. The suggested approach is applied to a modified IEEE 300-bus test system, and compared to the Monte Carlo simulation, where the DC-OPF is solved for each sample. Future work will study the relation between the proposed approach and automatic generation control, as well as extensions to the nonlinear AC-OPF problem.

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